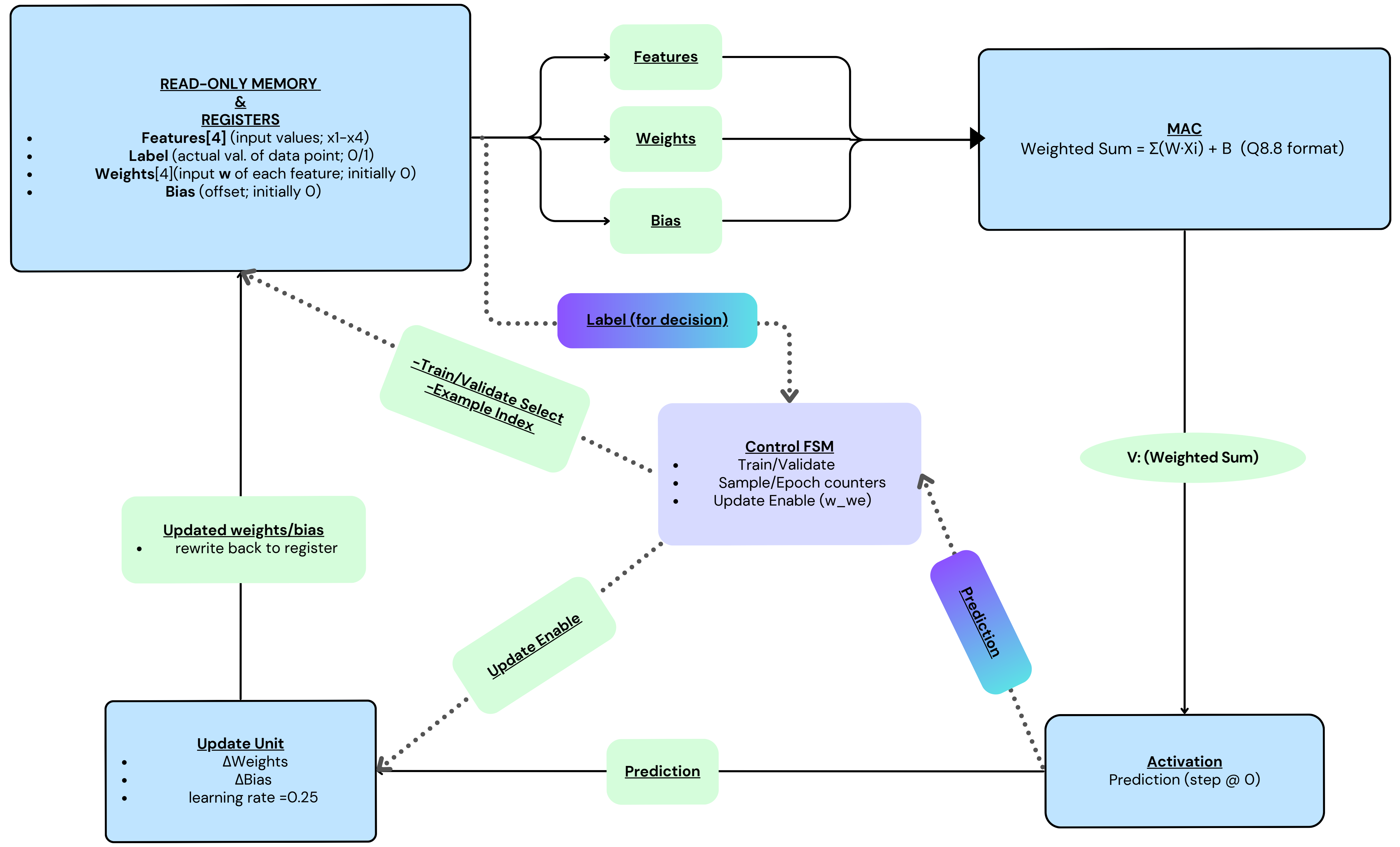
AI Project Milestone 1

By: Ahmed Elag & Marc Niven

Architecture



Update Rule

ERROR
e = y - y_hat ∈ {-1,0,+1}

PERCEPTRON
w <- w + (e/4)*x
b <- b + e/4
(skip if e=0)

Formats

X:	**WEIGHTS:**	**BIAS:**	**SUM:**
-Q4.4	-Q8.8	-Q8.8	-Q8.8

MAC:
- w(Q8.8) * x(Q4.4) ->Q12.12
- Accumulate in Q12.12
- Output shift >>4 -> Q8.8

EDA PLAYGROUND RUN STEPS

SETUP:
1) Site: edaplayground.com
2) Language: VHDL
3) Tools & Sim: Aldec Riveira Pro
4) Check: "Open EPWave after run"

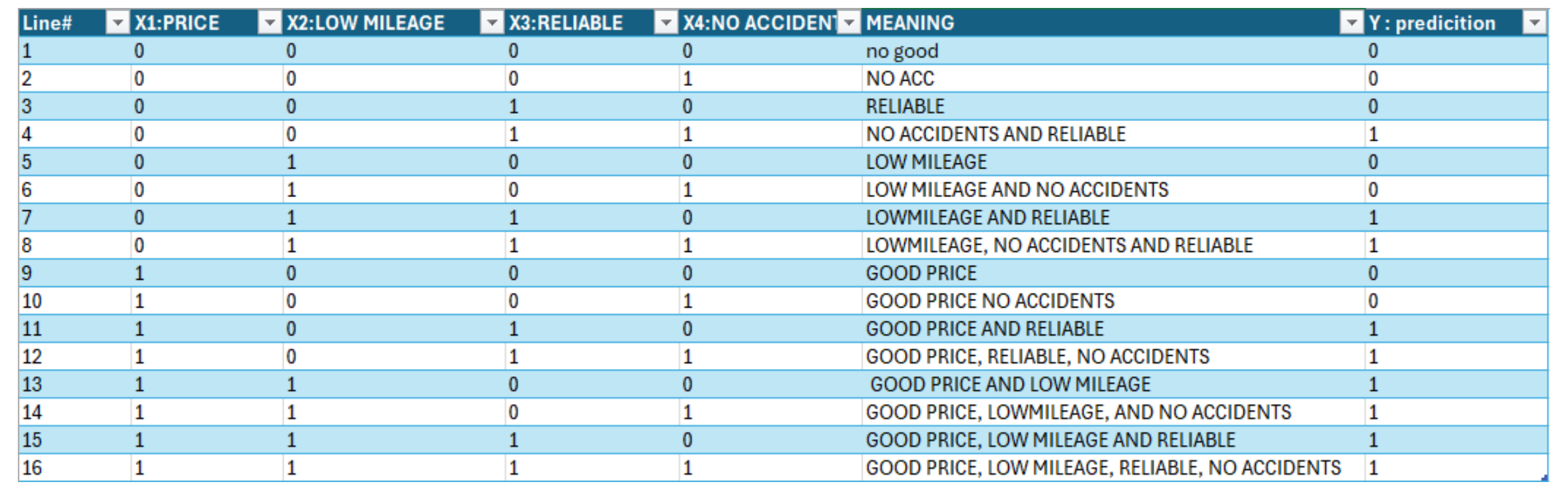
TOP MOD & GENERICS:
- Top entity to run: perceptron_top_tb
- If your DUT/testbench expose generics, set:
- N_TRAIN=6, N_VALID=3, MAX_EPOC=4 (we stop at epoch 4)

FILES:
- fx_types.vhd
- mac_unit.vhd
- act_unit.vhd
- update_unit.vhd
- control_fsm.vhd
- design.vhd(perceptron_core)
- testbench.vhd

Data

Data set

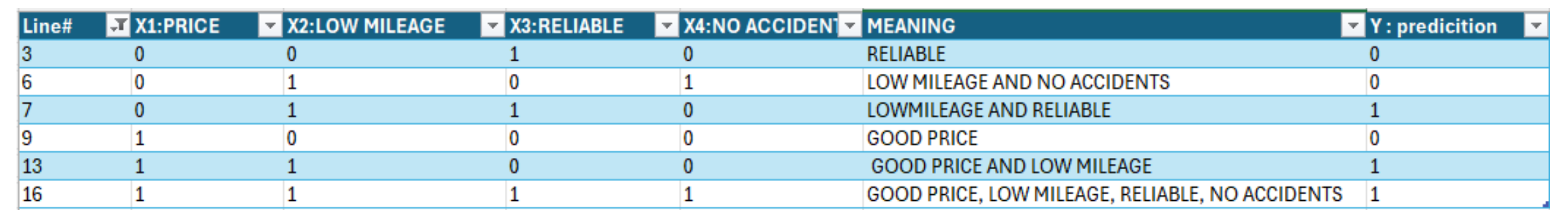
Prediction based on whether car would be sold on a market or not based on if it is:
- Well Priced
- Has Low Mileage
- Is Reliable
- Has no accidents



Line#	X1:PRICE	X2:LOW MILEAGE	X3:RELIABLE	X4:NO ACCIDEN	MEANING	Y : predicition
1	0	0	0	0	no good	0
2	0	0	0	1	NO ACC	0
3	0	0	1	0	RELIABLE	0
4	0	0	1	1	NO ACCIDENTS AND RELIABLE	1
5	0	1	0	0	LOW MILEAGE	0
6	0	1	0	1	LOW MILEAGE AND NO ACCIDENTS	0
7	0	1	1	0	LOWMILEAGE AND RELIABLE	1
8	0	1	1	1	LOWMILEAGE, NO ACCIDENTS AND RELIABLE	1
9	1	0	0	0	GOOD PRICE	0
10	1	0	0	1	GOOD PRICE NO ACCIDENTS	0
11	1	0	1	0	GOOD PRICE AND RELIABLE	1
12	1	0	1	1	GOOD PRICE, RELIABLE, NO ACCIDENTS	1
13	1	1	0	0	GOOD PRICE AND LOW MILEAGE	1
14	1	1	0	1	GOOD PRICE, LOWMILEAGE, AND NO ACCIDENTS	1
15	1	1	1	0	GOOD PRICE, LOW MILEAGE AND RELIABLE	1
16	1	1	1	1	GOOD PRICE, LOW MILEAGE, RELIABLE, NO ACCIDENTS	1

Training Data set

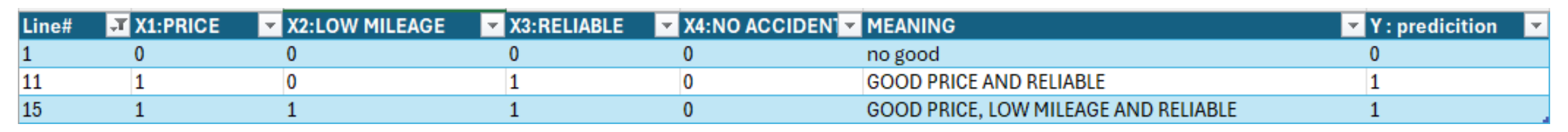
Chose the following 6 for training:



Line#	X1:PRICE	X2:LOW MILEAGE	X3:RELIABLE	X4:NO ACCIDEN	MEANING	Y : predicition
3	0	0	1	0	RELIABLE	0
6	0	1	0	1	LOW MILEAGE AND NO ACCIDENTS	0
7	0	1	1	0	LOWMILEAGE AND RELIABLE	1
9	1	0	0	0	GOOD PRICE	0
13	1	1	0	0	GOOD PRICE AND LOW MILEAGE	1
16	1	1	1	1	GOOD PRICE, LOW MILEAGE, RELIABLE, NO ACCIDENTS	1

Validation Data set

Chose the following 3 for validation:



Line#	X1:PRICE	X2:LOW MILEAGE	X3:RELIABLE	X4:NO ACCIDEN	MEANING	Y : predicition
1	0	0	0	0	no good	0
11	1	0	1	0	GOOD PRICE AND RELIABLE	1
15	1	1	1	0	GOOD PRICE, LOW MILEAGE AND RELIABLE	1

EP WAVES

STOP AT EPOCH = 4

The y_pred and y_true converge when epoch = 4.

Therefore, no need to continue and we can stop at epoch = 4

FINAL VALUES:

BIAS :
- b = -192

WEIGHTS:
1. w1 = 64
2. w2 = 128
3. w3 = 128
4. w4 = 0

CITATION

OpenAI. (2025, October 27). ChatGPT (GPT-5) [Large language model]. https://chat.openai.com/